Global Health Trends Extraction using NLP

Using Social Media Data for Public Health Insights

Team Members

Saadhna Soman — SE23UARI107

Manasa Reddy — SE23UARI106

Vallika Adari — SE23UARI130

Anvi Holani — SE23UARI012

Siddharth Varma — SE23UARI043

NLP · Social Media Analysis · Public Health · Data Visualization

Motivation

COVID-19 showed how fast misinformation & panic can spread

Social Media as Indicator

Twitter serves as real-time indicator of public sentiment during health crises

Early Signal Needs

Governments need early signals on public awareness, emotional response & emerging issues

NLP Solution

NLP enables automated extraction of trends from millions of tweets at scale

Social Media Data → **NLP Analysis** → **Public Health Insights**

Problem Statement

How can we extract meaningful global health trends from large-scale COVID-related tweets?

This project aims to:

Analyze public sentiment
toward COVID-19 through automated sentiment analysis of tweets

Detect health-related topics
frequently discussed in COVID-related tweets

Identify geo-specific mentions
using Named Entity Recognition (NER) to locate discussions

Visualize temporal trends
to understand evolving public concerns about COVID-19

Raw Data → NLP Processing → Insights

Related Work

Previous Studies

Epidemic Monitoring

Studies using Twitter data for monitoring Zika, Influenza, and COVID-19 outbreaks

Sentiment Analysis

VADER model widely used for social media analytics to gauge public sentiment

Named Entity Recognition

NER techniques used to identify outbreak locations and public health issues

Keyword Extraction

YAKE/Tf-idf methods used for topic identification in social media text

Our Contribution

Integrated NLP Pipeline

Our approach integrates all three methodologies into one comprehensive pipeline for trend extraction from COVID-related tweets

Key Innovations

- Combined sentiment analysis, NER, and keyword extraction
- End-to-end pipeline for automated trend extraction
- Temporal analysis of trends to understand evolving concerns

Dataset

Corona Tweets Classification NLP

Dataset from HuggingFace containing COVID-related tweets for analysis

Dataset Size

~41,000 COVID-related tweets

Global Coverage

Tweets from users worldwide during early pandemic stages

Dataset Columns

Tweet Text	Sentiment	Location	Date
Sample text content	Positive/Negative/Neutral	User location	Post date

Use Case Suitability

This dataset is suitable for sentiment analysis and topic trend identification in COVID-related social media data, enabling insights into public perception and concerns.

Methodology

Our NLP Pipeline

1. Data Loading & Preprocessing

Remove URLs, mentions, hashtags, noise

2. Sentiment Analysis

VADER model (compound polarity score → Positive/Negative/Neutral)

3. Keyword Extraction

YAKE to extract top 1–2 word topics

4. Named Entity Recognition

spaCy model to extract locations mentioned in tweets

5. Trend Analysis

Sentiment over time

Tweet volume

Keyword frequency

Experiments

Experimental Setup

Dataset
~41,000 COVID-related tweets

Sentiment Analysis
VADER applied to ~40K tweets

Keyword Extraction
5 keywords per tweet

Experiment Workflow

Data Loading → Preprocessing → Sentiment → Keywords → NER → Trend Analysis

Key Findings

Trend Visualizations

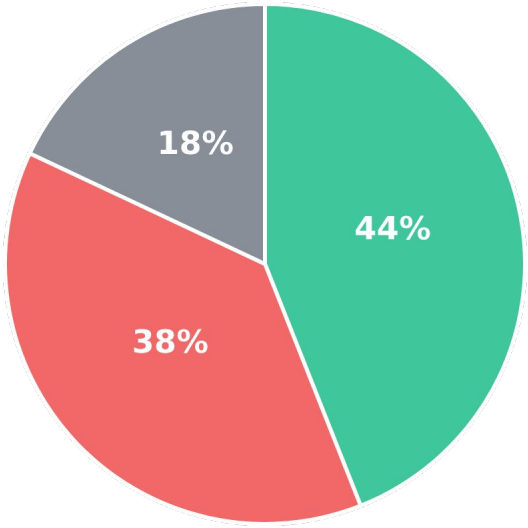
- Tweet volume over time
- Sentiment trend analysis
- Keyword frequency patterns
- Top locations talked about

NER Performance

- Applied on random subset (4K tweets)
- Identified geo-specific mentions
- Used spaCy model for location extraction

Results

Sentiment Distribution



Positive: 44% Negative: 38% Neutral: 18%

Top Keywords

grocery store
coronavirus
sanitizer
online shopping
toilet paper

Trend Insights

- Negative sentiment spikes align with lockdown announcements
- "Grocery store" & "sanitizer" surged early in pandemic
- Location mentions centered around USA, UK, India

Analysis and Discussion

Key Findings

Emotional Landscape

Mixed emotions expressed — fear, frustration, hope

Negative sentiment spikes align with lockdown announcements

Keyword Insights

Frequent keywords reveal real-world concerns

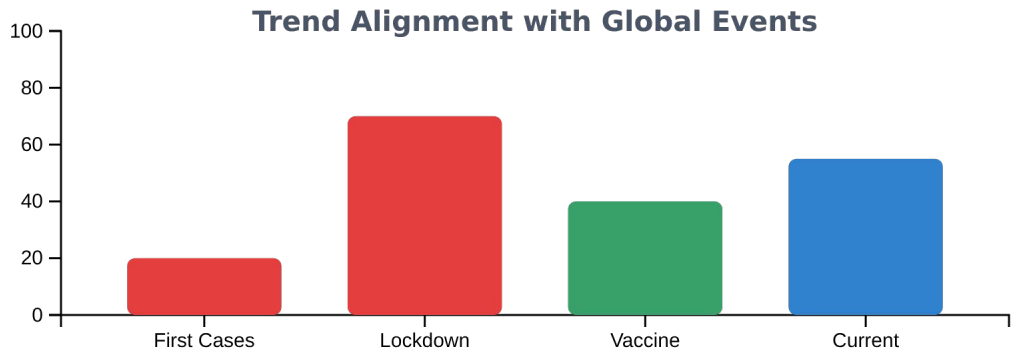
"Grocery store" & "sanitizer" surged early in pandemic

Geographical Concentration

NER shows discussions centered around USA, UK, India

Location mentions align with global pandemic hotspots

Implications for Public Health



Trend alignment with global events demonstrates NLP effectiveness

- NLP enables rapid, scalable analysis, unlike manual surveys
- Early detection of public fear or misinformation
- Supporting government decision-making with data

Contributions

What Our Team Built

Sentiment Analysis
VADER model for classifying emotion in tweets

Keyword Extraction
YAKE algorithm for identifying topics

Named Entity Recognition
spaCy model for extracting locations

Trend Visualizations
Graphs showing trends over time

Impact

- Clean, interpretable insights for health trend monitoring
- Reproducible code through Colab Notebook & GitHub

NLP Pipeline

Raw Tweets

↓

NLP Processing

↓

Health Insights

Project Timeline

Seven-week development schedule from literature review to final submission

Week 1
Literature review + dataset selection

Week 2
Data preprocessing + cleaning

Week 3
Sentiment analysis implementation

Week 4
Keyword extraction + NER

Week 5
Trend graphs + results

Week 6
Report writing + PPT preparation

Week 7
Final testing + submission

Development Approach

Iterative Development
Weekly sprints with integration of new components

Pipeline Integration
Combining sentiment analysis, keyword extraction, and NER

Evaluation
Testing on sample data before full implementation

Demo and Applications

Real-world Use Cases

Early Detection
Identify public fear or misinformation about health issues

Supply Monitoring
Track demand for essential supplies like masks and sanitizers

Regional Analysis
Understand location-specific COVID-related concerns

Policy Support
Inform government decision-making with data-driven insights

Demo Components

Sentiment Analysis
Visualize public sentiment trends from social media data

Keyword Extraction
Identify trending topics with live keyword analysis

NER Demo
Show location extraction from sample tweets

Colab Notebook
Interactive demonstration of the NLP pipeline

NLP-based health trend extraction enables rapid, scalable analysis for public health monitoring